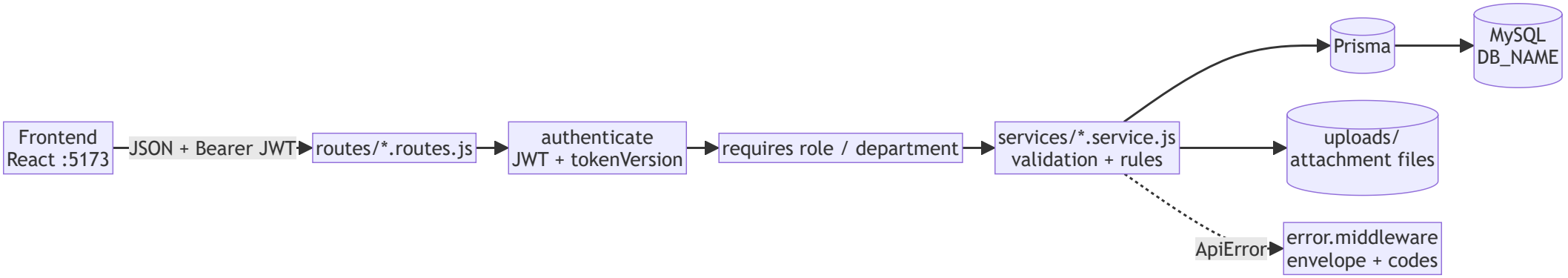
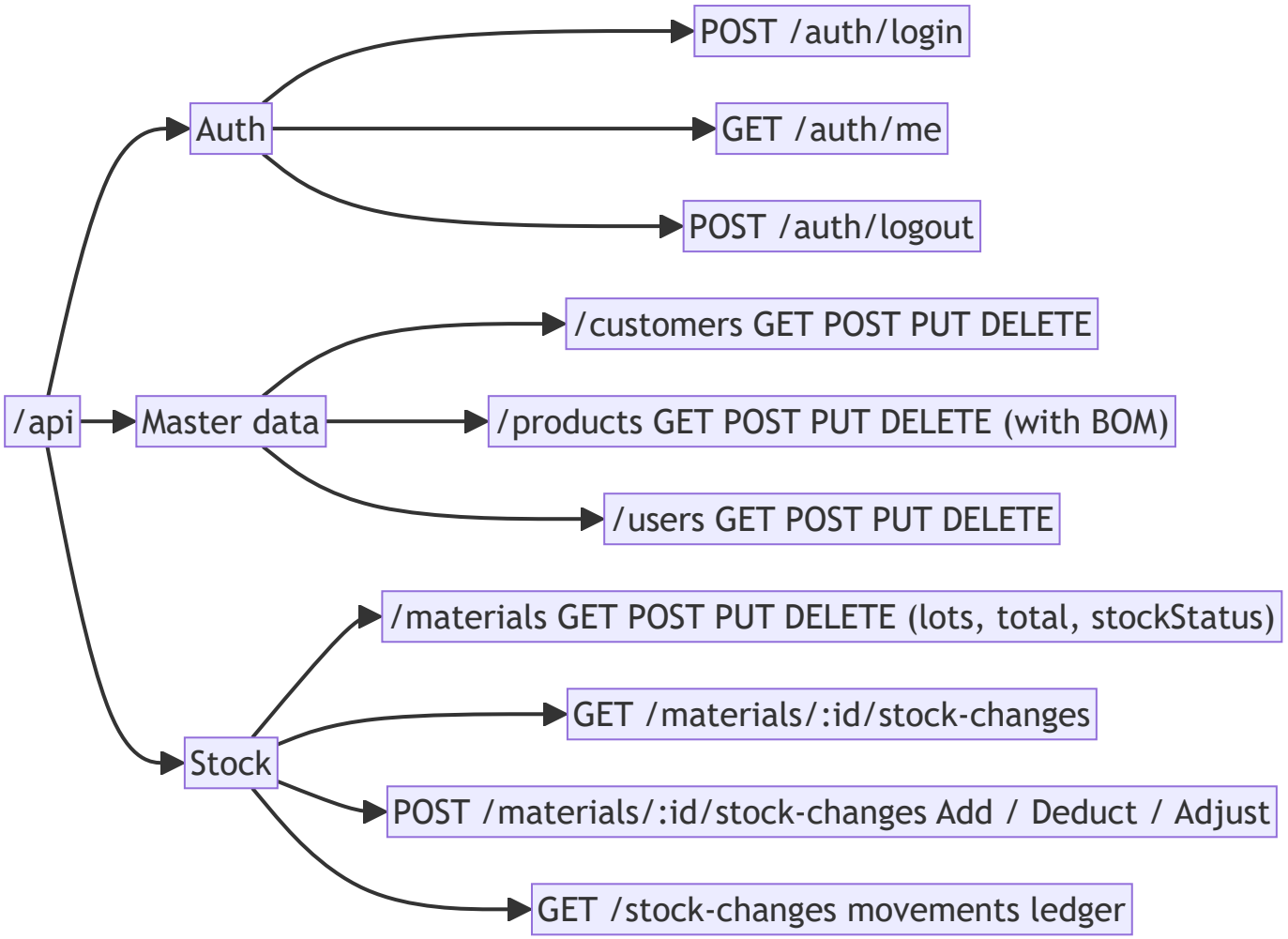
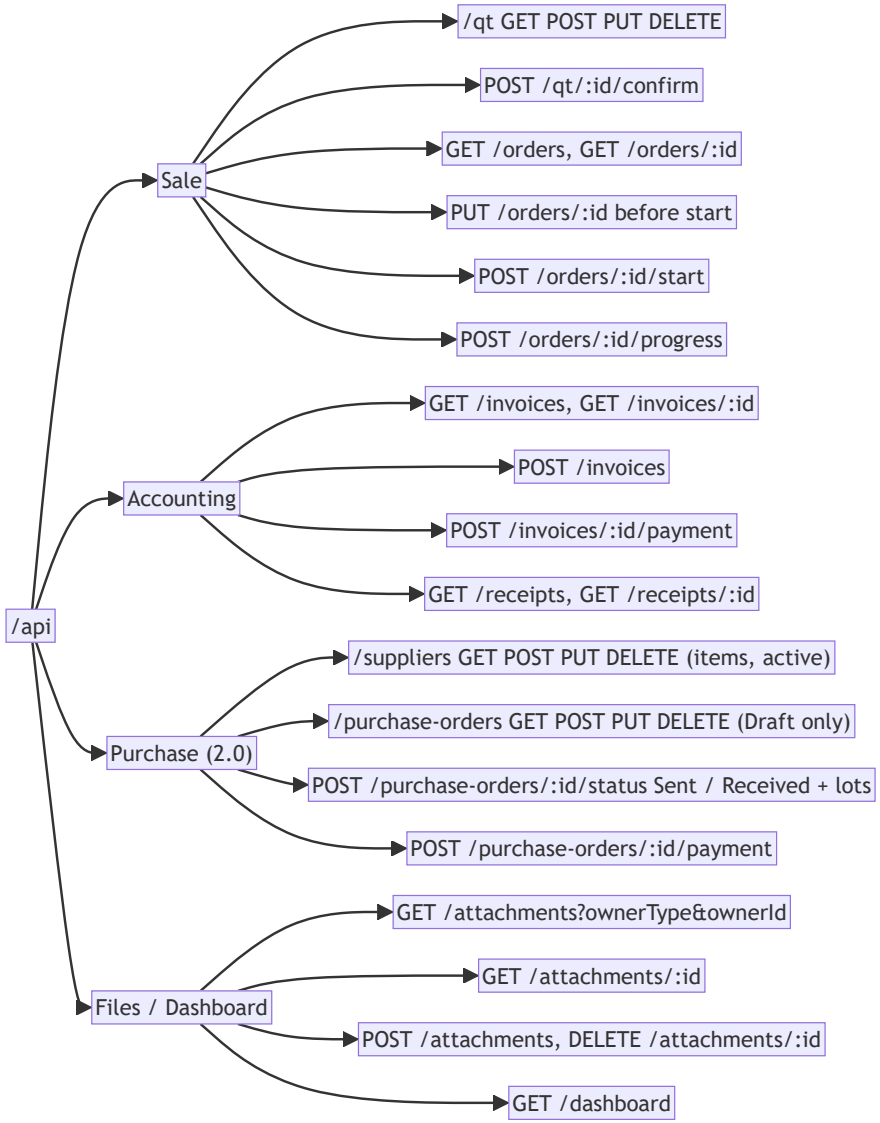
APIs Diagram

1. Layers



2. Endpoints by module





Query options added by the merged contract: `GET /qt` takes `step` (`Draft`, `Confirmed`, `InProduction`, `Done`) and every QT has a computed `step`; `GET /orders` takes `invoiced` (`true` / `false`); `GET /suppliers` takes `materialId` and `active`, and a list row has `itemCount` instead of `items`; a purchase order carries `supplierAddress` and its payment is `{ paymentDate, amount, method, referenceNo, note }`.

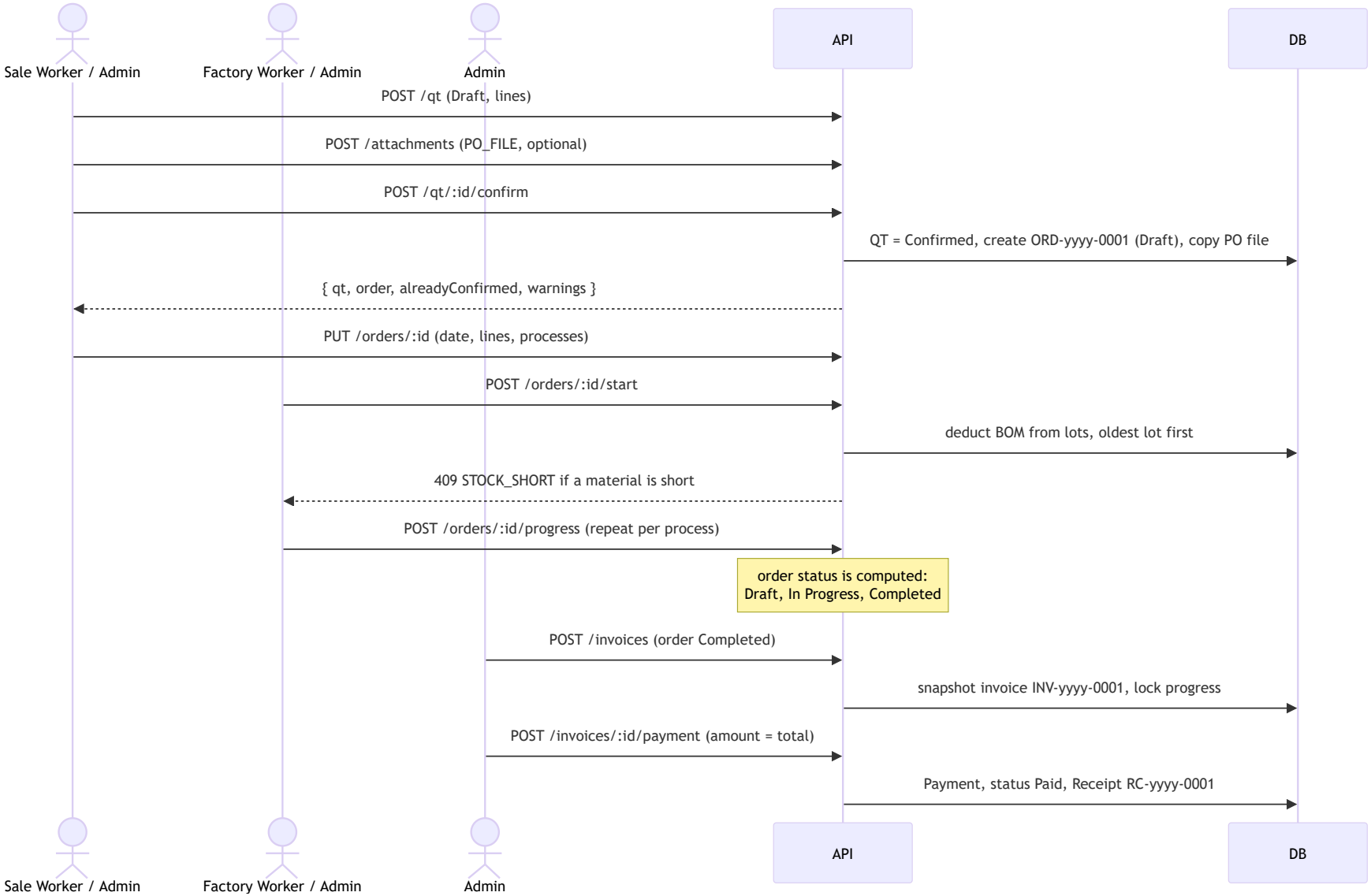
List calls take `q`, `page`, `pageSize` and answer `meta: { page, pageSize, total }`. `/orders`, `/qt`, `/invoices` also take `status`; `/materials` takes `status` = Normal / Low / Out; `/stock-changes` takes `type`, `from`, `to`.

3. Who may call

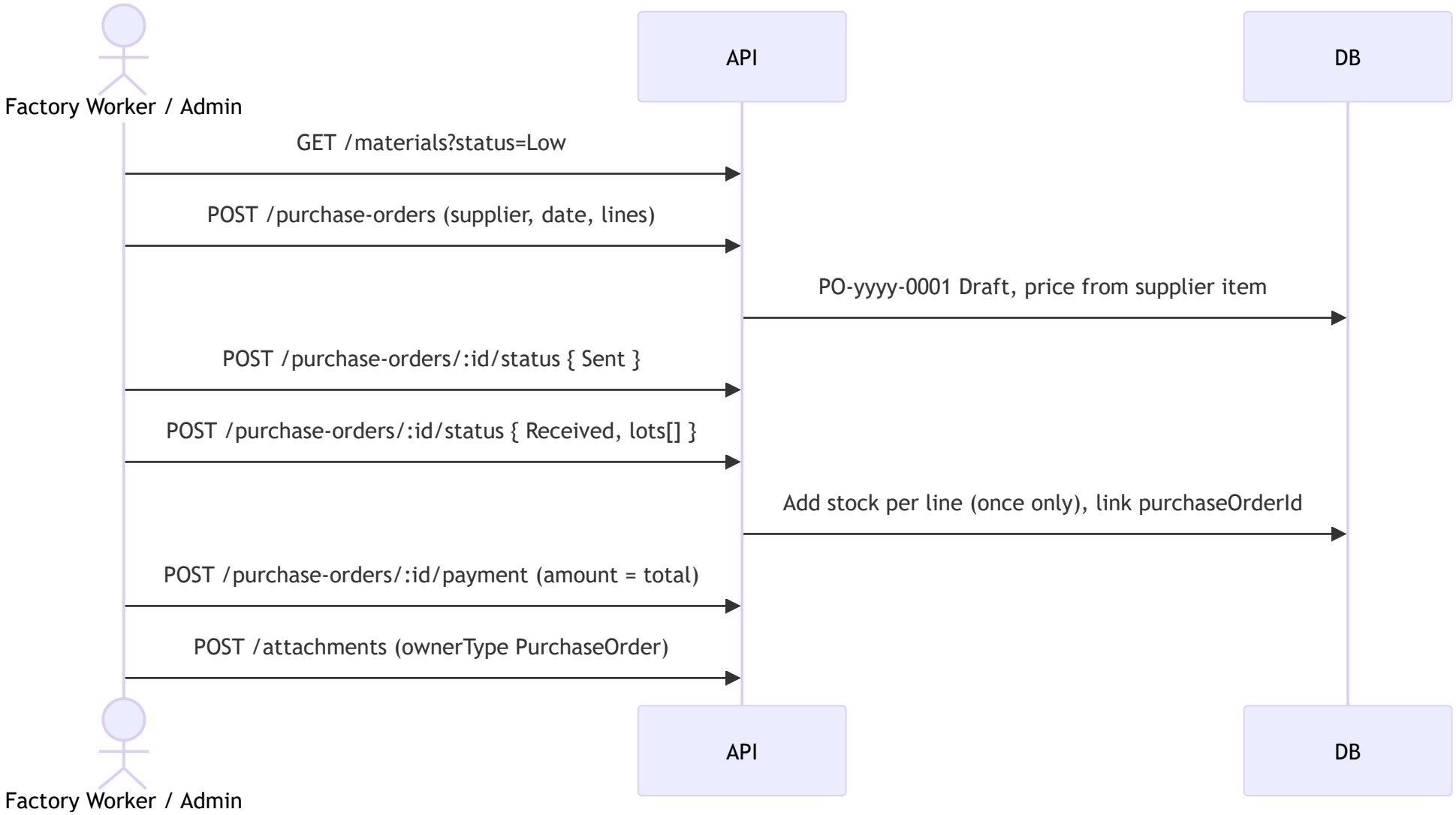
Area	Admin	Sale Worker (`Sales`)	Factory Worker (`Factory`)
Dashboard, products read, materials read, stock history, orders read	yes	yes	yes
Customers read	yes	yes	403
Customers, products, materials, users write	yes	403	403
Users read	yes	403	403
QT (all calls, Confirm, delete a Draft)	yes	yes	403
`PUT /orders/:id` (before start)	yes	yes	403
`POST /orders/:id/start`, `/progress`	yes	403	yes
Stock change by hand	yes	403	yes
Reopen / change a Completed process	yes	403	403
Suppliers read; purchase orders, payment, PO files	yes	403	yes
Suppliers write	yes	403	403
Invoices, payments, receipts	yes	403	403
Files of a QT (read, upload, delete)	yes	yes	403
Files of an order: read	yes	yes	yes
Files of an order: upload, delete	yes	403	yes

The role check runs before the record lookup, so a forbidden user gets `403`, not `404`.

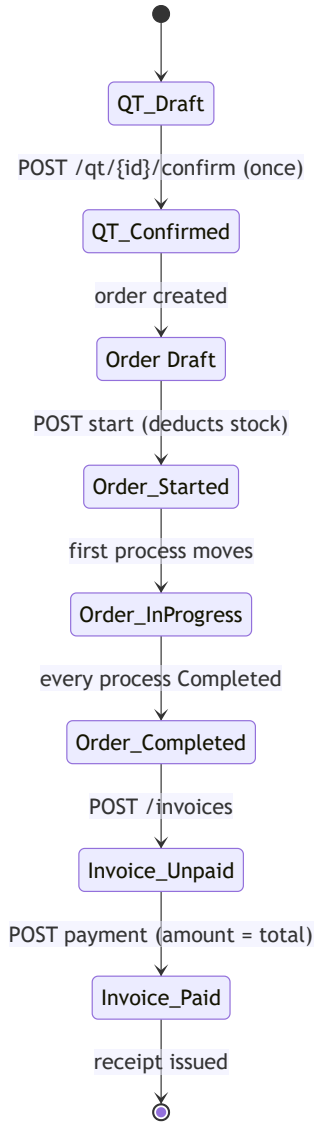
4. One sale, end to end



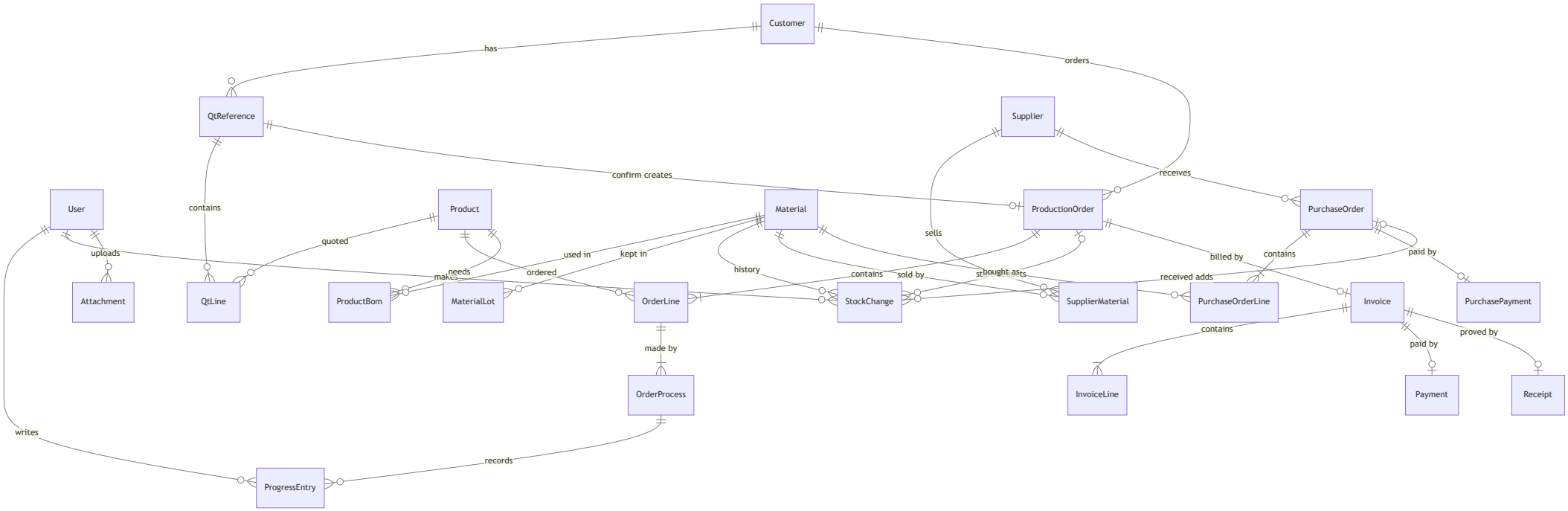
Purchase flow



5. Order and invoice states



6. Data model (tables)



`Attachment` points to its owner (`QtReference` or `ProductionOrder`) by `ownerType` + `ownerId`, with no database foreign key.

7. Error codes

HTTP	error	Typical cause
400	VALIDATION_ERROR	errors: [{ path, code }], codes REQUIRED, INVALID, INVALID_DATE, DUPLICATE, TOO_SHORT, TOO_HIGH, MUST_EQUAL_TOTAL, FILE_TYPE, FILE_TOO_BIG …
401	UNAUTHORIZED, INVALID_CREDENTIALS	No / bad / old token; wrong login
403	FORBIDDEN, COMPLETED_LOCKED	Role or department not allowed
404	NOT_FOUND	No such record
409	DUPLICATE_VALUE	Unique code, email, qtNumber used
409	IN_USE	Delete of a used record; errors: [{ kind, count, items }]
409	STOCK_SHORT, STOCK_NEGATIVE, LOT_NOT_FOUND	Stock rules
409	QT_LOCKED, ORDER_LOCKED, ALREADY_STARTED, ORDER_NOT_READY, ORDER_NOT_STARTED, ORDER_INVOICED	Order / QT lifecycle
409	PO_LOCKED, INVALID_STATUS_CHANGE, PO_NOT_RECEIVED	Purchase order lifecycle (ALREADY_PAID also covers a PO)
400	SUPPLIER_INACTIVE, NOT_SOLD_BY_SUPPLIER (codes inside VALIDATION_ERROR)	Purchase order lines
409	ORDER_NOT_COMPLETED, INVOICE_EXISTS, ALREADY_PAID, FILE_LOCKED	Accounting and files
429	TOO_MANY_REQUESTS	Login rate limit
500	INTERNAL_SERVER_ERROR	Unexpected failure

8. Tests

cd backend && pnpm test runs integration tests (Node test runner) that call the real Express app over HTTP against a separate database, DB_NAME_test (made from DATABASE_URL with _test added; the tests refuse to run on any other name, because they empty every table). The schema is pushed by test-support/setup-db.js first.

File	Covers
`test/auth.test.js`	login, `/me`, logout, sessions ended by a new password, rate limit, 401 on every module
`test/master.test.js`	customers, products with BOM, users: validation, search, paging, roles, `IN_USE`, duplicates
`test/stock.test.js`	materials, lots, status, stock changes (Add / Deduct / Adjust), history, movements, delete rules
`test/sale.test.js`	QT, Confirm QT, files (PO copy to the order), production orders, start (stock deduction), progress rules
`test/accounting.test.js`	invoices (snapshot), payment, receipts, dashboard
`test/purchase.test.js`	suppliers, purchase orders, status path, receiving (stock once), payment, files, in-use rules